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Monte-Carlo (I)

Exercice 1 (CDF inversion)**.**
Let F be a cumulative distribution function on $\mathbb{R}$. We define the generalized inverse of F (also known as quantile function), for any $u \in]0, 1[$:

$$F^{-1}(u) = \inf D_u = \inf\{x \in \mathbb{R} : F(x) \geq u\}.$$

1. Show that F^{-1} is non-decreasing.
2. Show that for all $x, F^{-1}(F(x)) \leq x$.
3. Show that D_u is closed and deduce that $F(F^{-1}(y)) \geq y$.
4. Conclude $F^{-1}(y) \leq x \Leftrightarrow y \leq F(x)$
5. Let F_X be the CDF of a random variable X. If $U \sim \mathcal{U}(0, 1)$, what is the distribution of $F_X^{-1}(U)$?
   Using the last question :

Exercice 2 (Reject sampling)**.**
We would like the generate samples from a distribution with density f given samples from a distribution with density g such that there exists $M > 0$ for which :

$$\forall x \qquad f(x) \leq Mg(x).$$

Consider the sampling algorithm :

1. generate X_i following density g and $U_I \sim \mathcal{U}[0, 1]$.
2. if $U_i \leq \alpha(X_i) = \frac{f(X_i)}{Mg(X_i)}$ then accept the sample X_i.

This algorithm (reject-sampling) can be formalized as follows. Let $(X_n, U_n)_{n\geq 1}$ be an i.i.d sequence where X_n has density g and independent of U_n which follows a uniform over $[0, 1]$. Let $T = \inf\{n, U_n \leq \alpha(Y_n)\}$ where $\alpha : x \mapsto \frac{f(x)}{Mg(x)}$.

1. Let $\varphi \in \mathcal{C}_b^+(\mathbb{R})$. Show that :

$$\mathbb{E}(\varphi(X_n)\mathbb{1}_{U_n \leq \alpha(X_n)})) = \frac{1}{M}\int_{\mathbb{R}^d} \varphi(x)f(x)\mathrm{d}x$$

2. What is then the probability of accepting a sample X_i ?
3. Compute $\mathbb{E}(\varphi(X_T))$ to show that X_T follows a distribution with density f.

Exercice 3 (Importance sampling : simple example)**.** We want to compute the probability that a Cauchy(0, 1) variable is larger than 2 using monte-carlo simulation.

$$p = \int_2^\infty \frac{1}{\pi(1 + x^2)}\mathrm{d}x = \frac{1}{2} - \frac{\arctan(2)}{\pi}$$

1. Using i.i.d samples $X_i \sim \mathcal{C}(0, 1)$, the crude monte-carlo estimator is given by :

$$\hat{p}_1 = \frac{1}{n}\sum_{i=1}^n \mathbb{1}(X_i \geq 2)$$

   What is the variance of $\hat{p}_1$? How can we decrease it using the same samples $(X_i)_i$?
2. The main drawback of $\hat{p}_1$ is the lack of samples at the tail of Cauchy distribution. Propose a different estimator $\hat{p}_2$ based on the following rewriting of p.

$$p = \frac{1}{2} - \int_0^2 \frac{1}{\pi(1 + x^2)}\mathrm{d}x$$

3. Using a change of variable, propose yet another estimator $\hat{p}_3$ so that the interval over which the integral is approximated is smaller.
4. Using integration by parts on the right function, compute analytically a primitive function of $x \mapsto \frac{1}{(1+x^2)^2}$.
5. Deduce the variance of the estimators and conclude.

Exercice 4 (Normalized Importance Sampling)**.** Consider a truncated scaled Gaussian random variable X on $[2, \infty]$. Its density is given by :

$$f(x) = \frac{\exp(-x^2/2)}{\int_2^\infty \exp(-x^2/2)} = \frac{\exp(-x^2/2)}{F} = \frac{\widetilde{f}(x)}{F}$$

We would like to estimate integrals of the form $\mathbb{E}_f(h(X))$ using an importance sampling proposal density g. The IS estimator is given :

$$\mathbb{E}_f(h(X)) = \int h(x)f(x)\mathrm{d}x = \int h(x)\frac{f(x)}{g(x)}g(x)\mathrm{d}x = \mathbb{E}_g(h(X)\frac{f(x)}{g(x)})$$

Let $w(X_i) = \frac{f(X_i)}{g(X_i)}$. The IS estimator is given by : $\frac{1}{n}\sum_{i=1}^n w(X_i)h(X_i)$ where X_i are iid samples from g.

1. Compute $\mathbb{E}(w(X_i))$.
2. Deduce an IS estimator for which we do not need to know the normalizing constants F and G.
3. Using a proposal density g of your choice, estimate the probability that the truncated Gaussian on $[2, +\infty]$ is larger than 4.

Exercice 5 (Control variates)**.** Let $X \sim \mathcal{N}(0, I_d)$. We would like to compute the probability of X belonging to a the set A :

$$A = \{(x_1, \ldots, x_d) | \prod_{i=1}^d |x_i| < C\}$$

denoted by p.

Let the crude monte-carlo estimator be denoted by $\hat{p}$ and consider an unbiased estimator $\hat{q}$ of some quantity q and define :

$$\hat{y} = \hat{p} + \alpha(\hat{q} - q).$$

1. Motivate the need for variance reduction in this case. How should we choose α ?

2. Propose a control variate estimator $\hat{q}$.